HW 6 Due 2/23

1) 8) 18 people 3 first, 2 last $3 \cdot 2! = 12$ combinations
18 people/pigeons > 12 combinations/pigeonholes
There are 6 more people than combinations, so at least 3 (6/2) share the same first/last name combination as someone else.

23-24 2) 23) Let $a_j < a_k$. If you take a subsequence (increasing) from a_k, let it be b_k. Say you add a_j to the beginning of it. Then the increasing subsequence b_j from a_j is $b_k + 1$. This results in the following inequality: $b_j \geq b_k + 1$.
$b_j \neq b_k$ so $(b_j, c_j) \neq (b_k, c_k)$. If $a_j > a_k$, $c_j \geq c_k + 1$ so $c_j \neq c_k$ so $(b_j, c_j) \neq (b_k, c_k)$. Thus, the ordered pairs are distinct, either $b_j \neq b_k$ or $c_j \neq c_k$. $b_j = b_k$ and $c_j = c_k$ are never both true when $j \neq k$.
24) As stated in the contradiction assumption, there are n choices for b_i and c_i. Therefore, there are $n \times n = n^2$ possible ordered pairs (b_i, c_i).

25) If $b_i > n$ or $c_i > n$, this would contradict our assumption that every increasing or decreasing subsequence has length n or less. b_i starts at a_i and c_i does as well. Therefore, b_i / c_i must at least contain a_i. Thus, $1 \leq b_i \leq n$ and $1 \leq c_i \leq n$.

26) The contradiction is that there are $n^2 + 1$ pigeons and n^2 holes. We have $n^2 + 1$ pairs $(b_1, c_1) \ldots (b_{n^2+1}, c_{n^2+1})$. Per 24, there are n^2 possible ordered pairs. $n^2 + 1 > n^2$, so at least one pair would not be distinct. $b_i = b_k$ or $c_i = c_k$. This is not possible as we proved in 23, so our assumption that every increasing or decreasing subsequence has length n or less is incorrect.

3) $s_i = \begin{cases} r_i & \text{if } r_i \leq 8 \\ 16 - r_i & \text{if } r_i > 8 \end{cases}$ $\quad r_i = a_i \bmod 16$

r_i will always be values 0-15 because of the nature of a mod function. The remainder of $a_i / 16$ can never be equal to or greater than 16. When $0 \leq r_i \leq 8$, $0 \leq s_i \leq 8$. When $r_i > 8$ (or $8 < r_i \leq 15$), $16 - 15 \leq s_i < 8 = 1 \leq s_i < 8$. Thus, $s_1 \ldots s_{10}$ range in value from 0 to 8.

28) There are 10 values in $a_1 \ldots a_{10}$. Say $j \neq k$ and $a_j \neq a_k$. There are only 9 possible values for any s_i (#27: 0-8). Thus, since $9 < 10$, or 9 holes < 10 pigeons, $s_j = s_k$ for some $j \neq k$.

29) suppose $s_j = s_k$ for some $j \neq k$. This means $r_j = r_k$ if $r_j, r_k \leq 8$. (given in problem) Or, $16 - r_j = 16 - r_k$ if $r_j, r_k > 8$. In either case, $r_j = r_k$ which means $a_j \bmod 16 = a_k \bmod 16$. Using the definition of the mod function, $a_j = 16n_1 + r_j$ $\quad a_k = 16n_2 + r_k$. $\quad a_j - a_k = 16n_1 + r_j - 16n_2 - r_k$. $\quad a_j - a_k = 16(n_1 - n_2)$. Thus, 16 divides $a_j - a_k$. ($r_j - r_k = 0$)

30) $(s_j \neq r_j$ or $s_k \neq r_k)$ and $(s_j \neq 16 - r_j$ or $s_k \neq 16 - r_k)$

For the conditions to fail means the above holds.

This means either ① $s_j = r_j = a_j \bmod 16$ and $s_k = 16 - r_k = 16 - a_k \bmod 16$, or ② $s_k = r_k = a_k \bmod 16$ and $s_j = 16 - r_j = 16 - a_j \bmod 16$

Case 1: $s_j = a_j \bmod 16$ $\quad s_k = 16 - a_k \bmod 16$ $\quad s_j = s_k$, so:

$a_j \bmod 16 = 16 - a_k \bmod 16$ $\quad$ also: $a_j = 16q_j + r_j$

$a_j \bmod 16 + a_k \bmod 16 = 16$ $\quad a_k = 16q_k + r_k$

$\boxed{r_j + r_k = 16}$

$a_j + a_k = 16q_j + r_j + 16q_k + r_k$

$a_j + a_k = 16(q_j + q_k + 1)$

Thus, 16 divides $a_j + a_k$ when the conditions fail.

4) 19) $s_1 = 2$ 0, 1 $s_2 = 3$ 01, 11, 10 $s_3 = 5$ 010, 110, 101, 011, 111

$s_n = s_{n-1} + s_{n-2}$	init. cond.:	$s_1 = 2$
		$s_2 = 3$

Base step: 20) $s_n = f_{n+2}$ $f_0 = 0$ $f_1 = 1$ $f_2 = 1$ $f_3 = 2$ $f_4 = 3$ $f_5 = 5$ $f_6 = 8$

$s_1 \stackrel{?}{=} f_3$ $s_2 \stackrel{?}{=} f_4$

✓ $2 = 2$ ✓ $3 = 3$

Induction: Assume for some $n \geq 1$, $s_n = f_{n+2}$ (from base step).

by induction:

① $s_{n+1} = f_{(n+1)+2}$

$s_{n+1} = f_{n+3}$

② $s_n = f_{n+2}$

$s_{n-1} = f_{n+1}$

③ $s_{n+1} = s_{n+1-1} + s_{n+1-2}$ ← by recurrence relation

$s_{n+1} = s_n + s_{n-1}$

④ Plugging in step ②, $s_{n+1} = f_{n+2} + f_{n+1}$.

$f_{n+3} = f_{n+2} + f_{n+1}$

THUS, $s_{n+1} = f_{n+3}$.

6) 2) NO 3) NO 5) NO 6) yes, order 2 8) yes, order 2

9) yes, order 2

7) 14) $a_n = 2^n a_{n-1}$; $a_0 = 1$

$a_{n-1} = 2^n 2^{n-1} a_{n-2}$

$= 2^n 2^{n-1} 2^{n-2} a_{n-3}$

$= 2^{n+n-1+n-2} a_{n-3}$

$2^n \cdot 2^{n-1} \cdots 2^{n-(n-1)} a_1$

$\frac{n(n+1)}{2}$

$2^{(1+\cdots+n)} a_0$

$2^{\frac{n(n+1)}{2}} a_0$

5) n elements in first set, k choices in second set, k^n possible functions in total. Subtract not onto functions from k^n for answer.

Non-onto functions: domain: i codomain: k-i

1 to k-1

C(k,i): number of possible ways to choose k-i elements from k.

$\times$ S(n,i): number of onto functions from n to k-i objects

$\sum_{i=1}^{k-1} C(k,i)\,S(n,i)$: non onto functions

$$k^n - \sum_{i=1}^{k-1} C(k,i)\,S(n,i)$$

8) $\sqrt{a_n} = \sqrt{a_{n-1}} + 2\sqrt{a_{n-2}}$ $b_n = b_{n-1} + 2b_{n-2}$ $a_0 = 1$ $a_1 = 1$

$r^2 = r + 2 \rightarrow 0$

$(r-2)(r+1) = 0$

$r = 2, -1$

$b_n = B(2)^n + D(-1)^n$

$b_0 = \sqrt{a_0} = 1 = B(2)^0 + D(-1)^0 = B + D$

$b_1 = \sqrt{a_1} = 1 = B(2)^1 + D(-1)^1 = 2B - D$

$2B - D = 1$

$+ \; B + D = 1$

$3B = 2$

$B = 2/3$

$D = 1/3$

$b_n = \frac{2}{3}(2)^n + \frac{1}{3}(-1)^n$

$\sqrt{a_n} = \frac{2}{3}(2)^n + \frac{1}{3}(-1)^n$

$a_n = \left(\frac{2}{3}(2)^n + \frac{1}{3}(-1)^n\right)^2$